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MEASUREMENTS OF RECENT GEOMAGNETIC SECULAR VARIATION IN SOUTHEASTERN AUSTRALIA AND THE QUESTION OF DIPOLE WOBBLE

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Archaeomagnetic and radiocarbon measurements on a series of ancient Aboriginal fireplaces and burnt tree-stump sites in southeastern Australia are presented, and an outline of changes in the local direction of the earth's magnetic field over the last 1500 years is offered.

An analysis of these new data from the southern hemisphere and published data for the northern hemisphere indicates that a large change occurred in the orientation of the geomagnetic dipolar axis between about 1400 and 1650 A.D.

1. Introduction

It was not until approximately 300 years ago that direct measurements of the earth's magnetic field achieved a global coverage good enough to permit spherical harmonic analysis. Since then, little change has occurred in the orientation of the main dipole axis; the northern geomagnetic pole has moved a total of only 6° of arc since 1650 A.D., from 82.7°N, 319°E [1] to its 1965.0 location of 78.5°N, 290°E [2].

Over periods of 10^4 yr or more, however, it is well-known that the time-averaged geomagnetic field appears to be very much like that of a hypothetical geocentric dipole co-axial with the earth's rotational axis. (Worldwide palaeomagnetic data also reveal a pattern of small but systematic departures from this model, with local time-averaged records generally giving slightly more upward-pointing directions in mid-latitudes, but this can be readily explained in terms of a long-term preferred zonal distibution of non-dipole features, similar to the instantaneous one exhibited by the present-day nondipole field [3].)

It is therefore clear that the dipolar axis must, in the long term, either remain very close to the rotational axis (in which case the present-day separation of 11.5° would be unusual) or change its orientation fairly rapidly and/or frequently (if an appreciable angular separation is commonplace). In either case, the way in which the axis actually moves about the geographic axis is a matter of considerable interest. Various suggestions have been made, including a counterclockwise quasi-hypotrochoidal motion with a period of 1500 yr [4,5], a smooth counterclockwise precession with a period of 25,800 yr [6], and a jerky random walk [7,3]; but measurements of the presumed dipole movement are hampered by the lack of simultaneous records from several parts of the world and by the fact that its global signal appears to be small in comparison with the signals from localized changes in the nondipole field.

In this paper, some new archaeomagnetic data on the secular variation in southeastern Australia during Recent times are presented. A simple method is then used to combine all available worldwide measurements for the period 1000 to 500 yr B.P., in an attempt to average out contributions from the nondipole field and determine the mean position of the geomagnetic pole during that period.

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TABLE 1

Some historic instrumental measurements of declination and inclination in the region of southeastern Australia

Name	Location		1965.0 values for		Historic measurement		Year A.D.	Observer	Reference
	lat. (°S)	long. (°E)	declination (°)	inclination (°)	declination (°)	inclination (°)			
At sea	42.4	142.1	10.7	–	00.0	–	1642	Tasman	Hansteen [18, table 1, p. 30]
At sea	55.0	132.2	01.5	–	00.1	–	1840–1841	Ross	Sabine [19, p. 227]
Adventure Bay	43.4	147.5	13.7	–	08.4 *	–	1773–1788	Bayley, Cook, Bligh	Hansteen [18, table 1, p. 30]
Tasmania	43.5	147.0	–	–72.7	–	–71.0 *	1787–1793	Cook, Bertrand, Dentrecasteaux	Hansteen [18, table 2, p. 35]
Tasmania	43.6	147.0	–	–72.8	–	–71.6 *	1792	de Rossel	Sabine [20, p. 5]
Hobart Magnetic Observatory	42.9	147.4	13.4	–72.2	10.4	–70.7	1840–1841	Ross	Sabine [19, pp. 227, 165]
Hobart Magnetic Observatory	42.9	147.4	13.4	–72.2	10.0	–70.6	1845–1846	Kay	Sabine [21, p. 636]
Julias Is.	38.5	144.7	10.8	–	02.8	–	1800	Grant	Hansteen [18, table 1, p. 31]
Melbourne	37.8	145.0	10.8	–68.4	08.5	–67.1	1860	–	Bauer [22, p. 46]
Melbourne	37.8	145.0	10.8	–68.4	08.1	–67.1	1880	–	Bauer [22, p. 46]
Melbourne	37.8	145.0	10.8	–68.4	07.9	–67.0	1890	–	Bauer [22, p. 46]
Botany Bay	34.0	151.4	11.8	–64.3	08.0	–67.0	1770	Cook	Hansteen [18, table 1, p. 30; table 2, p. 34]
Port Jackson	33.8	151.4	11.8		09.0	–	1770	Cook	Hansteen [18, table 1, p. 32]
Port Jackson	33.8	151.4	11.8		08.5	–	1788	Hunter	Hansteen [18, table 1, p. 32]
Port Jackson	33.8	151.4	11.8		11.2	–	1795	Broughton	Hansteen [18, table 1, p. 32]
Garden Is.	33.9	151.3	11.7	–64.2	09.9	–62.9 *	1841	Ross	Sabine [23, pp. 120, 122]

* Averaged value.

The 1965.0 values listed were scaled from worldwide charts published by the Royal Greenwich Observatory.

Declination values are east of true north and negative values of inclination are above the horizontal.

2. Historic observations near southeastern Australia

Instrumental measurements of the geomagnetic field were made in the Southern Ocean and along the coastlines of Tasmania, Victoria and New South Wales by several early explorers, and some typical results are listed in Table 1. They show that easterly declination values have steadily increased (by a total of about 4°) over the last 200 yr, and the 1642 A.D. observation by Tasman in the Southern Ocean may perhaps indicate that this trend commenced more than 300 yr ago. Inclination measurements, except that of Cook for Botany Bay, indicate a slight steepening (by about 1.5°) since 1770 A.D.

3. Localities and sampling

Numerous ancient fireplaces are preserved around the Willandra Lakes in western New South Wales. Archaeomagnetic and radiocarbon results have previously been reported for a series of fireplaces, found in stratified dune sediments deposited more than 25,000 yr ago when the lakes were active [8]. The area was also occupied during the Recent period, long after the lakes had finally dried out at around 16,000 yr B.P. and many fireplaces, made visible by modern erosion, are found on top of the ancient lake floors, shorelines and surrounding plains. A representative sample of such fireplaces, mostly well preserved and in a variety of situations, has been investigated.

In addition to the fireplaces two burnt tree-stump sites, exposed in ancient sediments cut by the present-day Murray River, were studied. Detailed descriptions and locations of the fireplaces and tree-stump sites are given in Barbetti [9] and Barbetti and Polach [10].

All except one of the fireplaces were of the oven type, with a number of baked clay-sand ovenstones (typically 10 cm across) arranged on top of a bed of sand, ash and charcoal. Where field evidence suggested that they were undisturbed, individual ovenstones were oriented using both sun and magnetic compasses, and charcoal for radiocarbon dating was collected only from underneath the ovenstones. The remaining fireplace (F52), a small mound of baked clay, was encased in polyurethane foam and collected as a single oriented sample. The tree-stump sites were excavated so that their bowl-shaped bases were exposed; samples were collected from an undisturbed edge of MR5, and all parts of MR3.

4. Ages

All of the fireplaces were found within the present-day zone of soil formation, and even their relative ages could not be determined on field evidence. The only site for which meaningful stratigraphic information was available was MR5, which came from a section of point-bar sediments containing several similar sites with stratigraphically consistent ages [11].

Radiocarbon measurements were made on charcoal from each fireplace and tree-stump, and the results are listed in Table 2. Certain fireplaces with both ^{14}C ages and archaeomagnetic directions which are indistinguishable have been grouped together.

All conventional ^{14}C ages contain a systematic error due to past fluctuations in the atmospheric ^{14}C concentration and corrections have therefore been applied, using the method and tables of Clark [12] to place confidence limits on the true ages. Calibrated ages, denoted by upper case letters B.P., are used in other parts of the text.

5. Remanent magnetism

At the laboratory, several cylindrical specimens, usually 28 mm diameter and length, were cut from each sample. Directions of magnetization for some groups of specimens were measured on an astatic magnetometer [13], some on a Schonstedt SSM-1 5-Hz spinner magnetometer and others on a Digico spinner [14]. All specimens were measured before and after thermal cleaning at approximately 100°C, and a few specimens were demagnetized at temperatures up to 500°C; no significant changes in direction were observed in the samples for which results are reported in this paper. Thermal demagnetization experiments were carried out with a furnace designed by Barbetti [15] in an automatically controlled field-free space described by McElhinny et al. [16].

Mean directions of magnetization for all samples are listed in Table 2 (values of α_{95} for individual samples are not given, but were generally between 2° and 4°). Considerable scatter between samples from

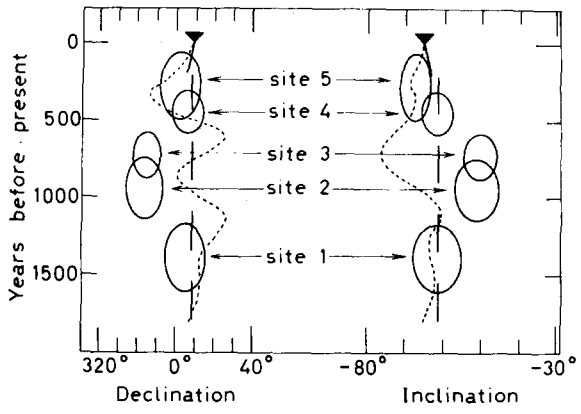


Fig. 1. Site mean declinations (0° to 360°, eastward from geographic north) and inclinations (negative, upwards) obtained from the fireplaces and one tree-stump, numbered as in Table 2. The ellipses enclose the areas within which the true values lie at the 95% confidence level. Solid arrows indicate the directions for 1965 A.D. and the short lines the historically observed trends. The vertical scale is in calibrated radiocarbon years B.P. (i.e. before 1950 A.D.). The dashed vertical lines give the values (9°, −61°) expected at the locality from the 1965.0 inclined dipole in the absence of the nondipole field. The dotted curves are hypothetical secular variation curves, calculated assuming that a steady 0.2°/yr westward drift of the 1965.0 nondipole field about the geographic axis (data from Yukutake and Tachinaka [41]) is superimposed on a fixed inclined dipole source identical to that for 1965.0.

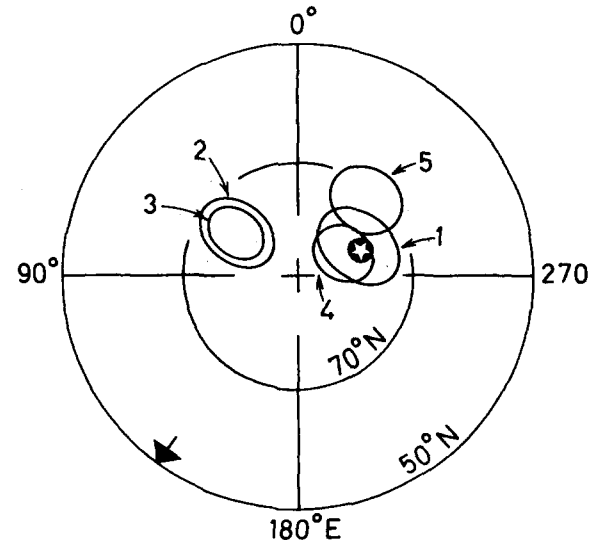


Fig. 2. Ellipses of 95% confidence about the north virtual geomagnetic pole (VGP) positions calculated from the site mean directions, numbered as in Table 2. The star indicates the pole position (78.5°N, 290°E) deduced by spherical harmonic analysis [2] for epoch 1965.0, the + the geographic pole, and the arrow the longitude (143°E) of the sampling localities. Two VGP locations (numbers 4 and 5) with ages less than 560 yr B.P. (and the VGP labelled number 1 at ~1400 yr B.P.) fall close to the 1965.0 pole, but the two pole positions (numbers 2 and 3) dated between 1135 and 565 yr B.P. are well separated and lie about 20° of arc to the east.

the same fireplace is evident in some cases (e.g., GF1), and is thought to be due to limited post-firing movement of ovenstones, perhaps caused by natural settling or external interference. An iterative but arbitrary procedure was therefore adopted, in which differences of 10° or more from an overall fireplace mean were taken to indicate a significantly different sample direction; that sample was then excluded, the mean recalculated, and the process repeated until all remaining sample directions were within 10° of the mean. Samples with directions considered to be aberrant are identified in Table 2.

The same procedure was also used when sample mean directions, from different fireplaces with similar directions of magnetization and overlapping ^{14}C ages, were grouped as indicated in Table 2. Some of the site mean values, particularly those dating between 1135 and 565 yr B.P., are derived from measurements on two or three fireplaces located a few tens of kilometers apart.

The five numbered site mean directions and appropriate calibrated ^{14}C ages are illustrated in Fig. 1. They provide a reasonably continuous outline of directional changes over the last 1500 yr, the most prominent feature of which is an apparently abrupt change at about 560 yr B.P. Corresponding virtual geomagnetic pole (VGP) positions are also listed in Table 2 and illustrated in Fig. 2, and it can be seen that those for the last 560 yr lie close to the present-day geomagnetic pole and away from the geographic pole, while those for the period 1135 to 565 yr B.P. lie elsewhere.

6. Discussion

An attempt was made to predict the local variation of the geomagnetic field over the last 1800 yr in the absence of dipole movement, by assuming that all variations might have been due to westward drift of

TABLE 2

Ages, and directions of magnetization after thermal cleaning at 100°C

Fire-place	^{14}C laboratory number	Conventional ^{14}C age (yr b.p.)	95% confidence limits on true age (yr B.P.)	Sample	n	Declination (°)	Inclination (°)	k	α_{95} (°)	Position of the north VGP
MR5	ANU-692	6140 ± 160	6580–7340	A	8	14.1	−50.1			
				B	7	6.4	−51.5			
				Mean:		10.3	−50.9			
F52	ANU-669	4260 ± 190	4350–5445	A	6	334.6	−55.5			
F4	ANU-665	4020 ± 100	4235–4925	B	2	345.2	−48.3			
UF4	ANU-679	3270 ± 90	3340–3835	A	6	21.0	−46.4	} directions significantly different		
				B		12.9	−62.1			
GF4	ANU-678	2440 ± 80	2350–2815	A	8	352.6	−50.8			
				B	7	352.6	−56.3			
				C	7	359.9	−51.3			
				Mean:		355.1	−52.8			
GF3	ANU-655	2010 ± 100	1725–2340	A	1	29.2	−57.4			
GF1	ANU-653	1420 ± 100	1185–1600	A	2	14.2	−62.2			
				B	3	12.2	−62.5			
				C	4	358.7	−57.4			
				E	1	7.8	−68.2			
				F	2	1.8	−57.0			
				Site mean (1):		6.4	−61.5	210 (N = 5, R = 4.9809)	5.2	79°N, 296°E (dp = 6.2°, dm = 8.0°)
				D	3	1.7	−43.6	} directions significantly different		
				G	1	331.4	−64.9			
UF1	ANU-674	990 ± 70	740–1135	A	5	336.4	−57.7			
				B	11	339.7	−49.6			
				C	1	351.6	−41.7			
				D	7	344.9	−55.3			
				E	6	342.2	−43.5			
F2	ANU-663	950 ± 120		A	1	352.3	−50.5			
				B	4	341.4	−57.2			
				Site mean (2):		344.4	−50.9	119 (N = 7, R = 6.9495)	5.5	77°N, 57°E (dp = 5.0°, dm = 7.4°)

TABLE 2 (continued)

Fire-place	^{14}C laboratory number	Conventional ^{14}C age (yr b.p.)	95% confidence limits on true age (yr B.P.)	Sample	n	Declination (°)	Inclination (°)	k	α_{95} (°)	Position of the north VGP
MB3	ANU-660	940 ± 50	730–1045	A	1	23.4	−53.0	directions significantly different		
				B	4	357.6	−30.0			
MB2	ANU-659	760 ± 150	565–885	A	4	342.0	−50.7			
				B	3	347.4	−45.7			
UF2	ANU-675	750 ± 70		B	3	348.3	−42.9			
				C	2	341.6	−55.0			
				D	2	340.3	−52.2			
				F	2	352.8	−56.0			
				G	2	341.9	−42.7			
F	ANU-666	740 ± 70		A	2	353.7	−54.3			
				Site mean (3):		345.8	−50.0	162 ($N = 8$, $R = 7.9569$)	4.3	78°N, 61°E ($dp = 3.8°$, $dm = 5.8°$)
				B	3	12.1	−49.1	direction significantly different		
MR3	ANU-691	400 ± 70	315–560	B	3	23.5	−68.3			
				C	9	352.0	−62.4			
				D	9	359.2	−64.1			
				E	6	6.7	−58.3			
				F	3	8.3	−55.3			
				H	7	14.1	−66.6			
				I	6	7.1	−61.3			
				J	6	2.5	−56.3			
				K	8	8.5	−61.2			
				Site mean (4):		6.3	−61.7	188 ($N = 9$, $R = 8.9574$)	3.7	81°N, 293°E ($dp = 4.4°$, $dm = 5.7°$)
AL3	ANU-672	240 ± 60	50–470	A	1	341.4	−68.0			
GF2	ANU-654	220 ± 60		B	2	5.0	−68.5			
				C	2	359.0	−67.0			
				E	2	16.6	−67.2			
				F	5	5.4	−68.0			
				G	1	4.7	−64.2			
				Site mean (5):		2.1	−67.4	299 ($N = 6$, $R = 5.9833$)	3.8	73°N, 318°E ($dp = 5.3°$, $dm = 6.3°$)

TABLE 2 (continued)

Fire-place	^{14}C laboratory number	Conventional ^{14}C age (yr b.p.)	95% confidence limits on true age (yr B.P.)	Sample	n	Declination (°)	Inclination (°)		k	α_{95} (°)	Position of the north VGP
				A	2	270.8	−76.4	}	directions significantly different		
				D	2	350.2	−79.4				
CM1	ANU-651	>modern	0–210	A	3	2.3	−63.6				
				B	5	0.8	−65.7				
					Mean:	1.6	−64.7				

Conventional radiocarbon ages, obtained from charcoal and based on a 5568-yr half-life, are listed with their standard errors; details of measurements have been published by Barbetti and Polach [10]. Where fireplaces are grouped, error-weighted mean ages were calculated, and the calibration tables of Clark [12] were used to place confidence intervals on the true ages (column 4). Magnetic measurements were made on n specimens from each oriented sample of baked clay-sand, and mean declinations (east of true north) and inclinations (negative upwards) are listed in columns 7 and 8. Mean directions are also given for groups of samples as indicated, and a numbered site mean is assigned where the number of samples (N) exceeds three. R is the length of the resultant of N unit vectors, k the Fisher [24] precision parameter and α_{95} the radius of the circle of 95% confidence about the site mean direction; dp and dm are the uncertainties (95% confidence level) in pole position, along and perpendicular to the great circle passing through the point of observation, respectively.

TABLE 3

Analysis of global VGP data for the period 950 A.D. to 1450 A.D.

Division	Country	Reference	Approximate mean coordinates of observations	
1N	Georgia, U.S.S.R.	Nodia et al. [25]	42°N, 44°E	
	Greece	Belshé et al. [26]	38°N, 24°E	
	Iceland	Brynjolfsson [27]	ca. 63°N, 20°W	
	Italy	Tanguy [28]	38°N, 15°E	
	Ukraine/Crimea	Burlatskaya et al. [29]	47°N, 32°E	
	Ukraine/Moldavia	Rusakov and Zagniy [30]	48°N, 29°E	
	United Kingdom *	Aitken and Weaver [31]	53°N, 1°W	
	United Kingdom	Aitken et al. [32]	53°N, 1°W	
	United Kingdom	Aitken and Hawley [33]	53°N, 1°W	
	United Kingdom	Aitken and Hawley [34]	53°N, 1°W	
				100-yr mean VGP:
1S				
2N	Japan	Hirooka [5]	35°N, 136°E	
	Japan	Kawai et al. [35] **	35°N, 136°E	
	Japan	Yukutake et al. [36]	35°N, 136°E	
				100-yr mean VGP:
2S	Australia	this paper	33°S, 143°E	
				VGP:
3N				
3S				
4N	U.S.A. (southwest)	Dubois [37]	32°N, 111°W	
				VGP ***:
4S				

* Only data classified with "A" or "B" reliability by the original authors have been included in this analysis.

** Contains a list of sites and graph of results, for which twenty values are tabulated by Kinoshita [38]; thirteen of these results are also tabulated by Hirooka [5] and included in the preceding entry; they are not duplicated here. The three results from Oshima volcano, Izu, were averaged with those of Yukutake et al. [36] and are included in the following entry. (Kinoshita [38] also attributes six results for this period to Kawai et al. [39,40], but they are merely preliminary results for which final values are given in this reference [35].)

*** Values for 1000, 1100, 1200, 1300 and 1400 A.D. scaled from Figs. 1 and 2 of cited reference.

The arbitrary divisions used are north (N) or south (S) of the equator, and lie between the meridians 330°E and 60°E (divisions 1N and 1S), 60°E and 150°E (2N and 2S), 150°E and 240°E (3N and 3S), and 240°E and 330°E (4N and 4S).

n gives the numbers of independent VGP determinations, *N* the total included in calculations of the 100-yr mean using Fisher [24] statistics, and *R* the resultant of *N* unit vectors representing the VGP.

an otherwise unchanging nondipole field. The results are illustrated in Fig. 1, and the inadequacy of this particular model in terms of fine detail is obvious from its failure to predict even the historically observed trend towards steeper inclination over the last 200 yr. Nevertheless an overall behaviour similar to that generally found by Cox [3], in which predominantly more upwards-pointing directions are produced in mid-latitudes by adding the nondipole field to a dipole field, is exhibited. In view of this property of the nondipole field the archaeomagnetic directions observed between 1135 and 565 yr B.P. are abnormally shallow;

n				
950–1049 A.D.	1050–1149 A.D.	1150–1249 A.D.	1250–1349 A.D.	1350–1450 A.D.
1	1	1	–	3
–	–	–	1	–
2	–	2	1	1
–	–	–	1	1
–	1	1	–	–
3	4	2	–	7
2	2	–	3	–
–	–	1	1	–
3	1	1	3	1
–	–	1	2	2
79°N, 79°E (N = 11, R = 10.9302)	78°N, 119°E (N = 9, R = 8.9357)	80°N, 132°E (N = 9, R = 8.9702)	76°N, 164°E (N = 12, R = 11.8597)	83°N, 172°E (N = 15, R = 14.7800)
no data				
1	3	2	7	6
2	–	–	2	–
1	1	–	1	1
77°N, 25°E (N = 4, R = 3.9765)	86°N, 339°E (N = 4, R = 3.9826)	86°N, 122°E (N = 2, R = 1.9712)	81°N, 226°E (N = 10, R = 9.8899)	83°N, 283°E (N = 7, R = 6.9278)
1	–	1	–	–
77°N, 57°E		78°N, 61°E		
no data				
no data				
1	1	1	1	1
88°N, 63°E	84°N, 211°E	78°N, 193°E	85°N, 164°E	85°N, 254°E
no data				

they would require an ancient nondipole field with large downwards- and westwards-pointing components, in sharp contrast with mid-latitude values for the present-day nondipole field. A different orientation of the dipole axis seems a more likely explanation.

If dipole movement is the correct explanation, then a coherent signal must be present in records of the secular variation from other parts of the world. A summary of all published archaeomagnetic data known to the author for the period 950 A.D. to 1450 A.D. is presented in Table 3. The surface of the earth was arbitrarily divided into eight equal-area spherical tri-

angles, VGP data from within these divisions were grouped into 100 yr class intervals and mean VGP locations were calculated. This procedure would effect limited space- and time-averaging of the nondipole field. The northern hemisphere results resemble those compiled by Kawai and Hirooka [4] and Hirooka [5], with some broad similarities as well as considerable differences in the apparent VGP paths; but the Australian data indicate a stationary VGP in a position quite different from most of the northern hemisphere VGP. Those differences make it clear that any coherent signal arising from dipole movement within that 500-yr period must be small in comparison with most regional nondipole signals.

Given the need to further average the nondipole signal with the limited data currently available, the procedure adopted was to calculate time-averaged 500-yr mean VGP locations for each division. The results are listed in Table 4 and illustrated in Fig. 3, and a reasonable grouping is evident. The 500-yr division means were then combined to give a time- and space-averaged grand mean VGP, located 15.5° from the 1965.0 geomagnetic pole. The average orientation of the geomagnetic dipole axis for the period 950 A.D. to 1450 A.D. must therefore have been quite different and well-separated from the orientations observed since 1650 A.D.

TABLE 4

Mean VGP for the period 950 A.D. to 1450 A.D. observed from different divisions of the earth's surface

Division	Mean VGP		N	R	K	A_{95} (°)
	(°N)	(°E)				
1N	81	132	5	4.9707	137	6.5
2N	88	323	5	4.9523	84	8.4
2S	78	59	2	1.9999		
4N	85	200	5	4.9815	217	5.2
Grand mean:	86	100	4	3.9676	93	9.5

Mean VGP locations were calculated for each division from N values of the 100-yr mean VGP (obtained from Table 3). R is the resultant of N unit vectors representing the VGP, K is the Fisher [24] precision parameter and A_{95} the radius of the circle of 95% confidence about the mean. The grand mean was calculated from the four division means.

The above result may be compared (Fig. 3) with that obtained by Doell [17] from the Kau volcanic series, where 54 superimposed lava flows gave VGP locations that are closely grouped together and also well-separated from the 1965.0 geomagnetic pole. Doell and Cox [7] concluded that the lava flows were erupted over a period of time much longer than 500 yr when the local nondipole field was small, sometime between 10,000 yr and a few hundred years B.P. Given the similarity of the Kau VGP locations and the archaeomagnetic grand mean VGP, it is tempting to speculate that the geomagnetic dipole axis may have been nearly stationary for a period of 10^3 or more years, ending at about 1400 A.D. But regardless of whether or not it was stationary before 1400 A.D., there can now be little doubt that the dipole axis moved rapidly after that date and by 1650 A.D. had reached a position very close to that which it occupies at the present time.

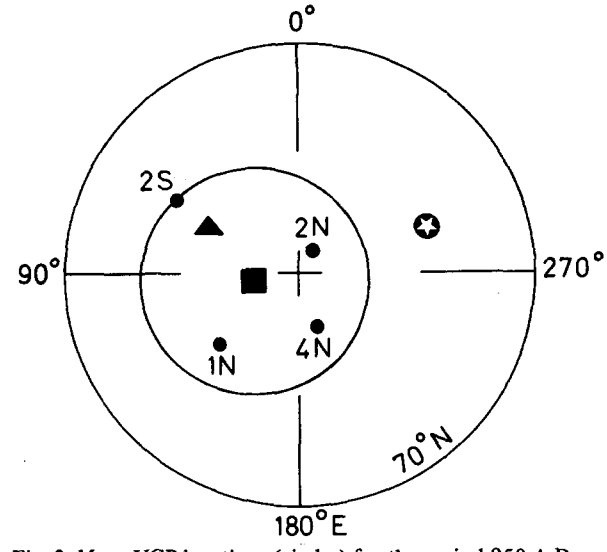


Fig. 3. Mean VGP locations (circles) for the period 950 A.D. to 1450 A.D., as observed from the four divisions of the earth's surface for which measurements are available. The divisions are those used in Table 3 and the mean poles those listed in Table 4; while the square and associated circle of 95% confidence is the grand mean of those four VGP. The star indicates the geomagnetic pole position (78.5°N, 290°E) for 1965.0, the + the geographic pole, and the triangle is the mean VGP given by Doell [17] for the Kau volcanic series in Hawaii (which lies in division 3N and is dated to a period of time somewhere between 10,000 yr and a few hundred years B.P.).

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